LIFTING INVERTIBLE SHEAVES FROM A CLOSED SUBSCHEME IN DIMENSION AT MOST ONE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Vanishing of higher cohomology of abelian sheaves on a Noetherian space of dimension at most one. Stacks Project, Cohomology, Proposition cohomology-proposition-vanishing-Noetherian (tag 02UU). Kills H^2 and hence the obstruction to lifting unit classes.
- Cohomology of a closed immersion and the Picard group as H^1 of the unit sheaf. Stacks Project, Cohomology, Lemma cohomology-lemma-cohomology-and-closed-immersions (tag 02UV) and Lemma cohomology-lemma-h1-invertible (tag 09NU). Translate the two H^1 groups into groups of invertible sheaves on Z and X.

Source reference. Original reference 02UZ: sources/stacks-chow/tags.txt, source lines 3750–3750. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02UV: sources/stacks-chow/tags.txt, source lines 3746–3746. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 09NU: sources/stacks-chow/tags.txt, source lines 12227–12227. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $i : Z \to X$ be a closed immersion of schemes. If the underlying topological space of X is Noetherian and $\dim(X) \leq 1$, then $\mathrm{Pic}(X) \to \mathrm{Pic}(Z)$ is surjective.*

Proof. Consider the short exact sequence

$$0 \to (1 + \mathcal{I}) \cap \mathcal{O}_X^* \to \mathcal{O}_X^* \to i_* \mathcal{O}_Z^* \to 0$$

of sheaves of abelian groups on X where $\mathcal{I}$ is the quasi-coherent sheaf of ideals corresponding to Z. Since $\dim(X) \leq 1$ we see that $H^2(X, \mathcal{F}) = 0$ for any abelian sheaf $\mathcal{F}$, see Cohomology, Proposition 02UZ. Hence the map $H^1(X, \mathcal{O}_X^*) \to H^1(X, i_* \mathcal{O}_Z^*)$ is surjective. By Cohomology, Lemma 02UV we have $H^1(X, i_* \mathcal{O}_Z^*) = H^1(Z, \mathcal{O}_Z^*)$. This proves the lemma by Cohomology, Lemma 09NU. □

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